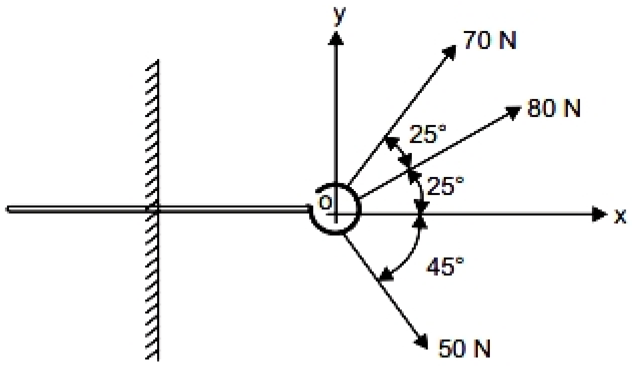
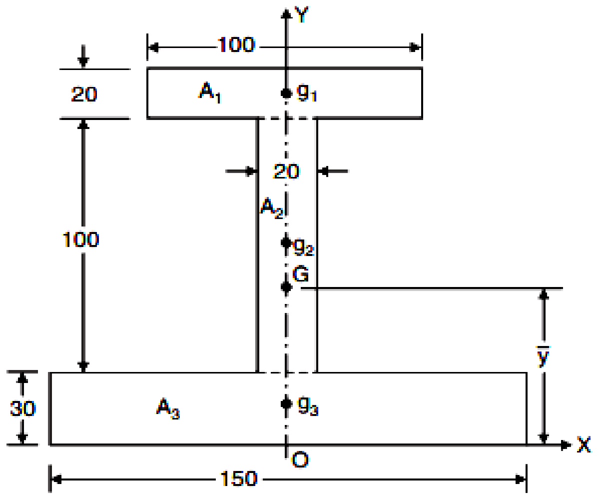
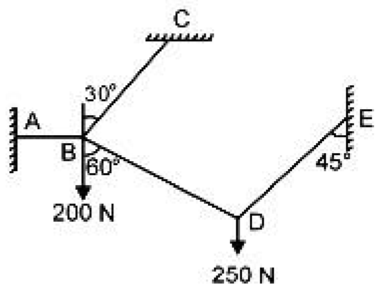
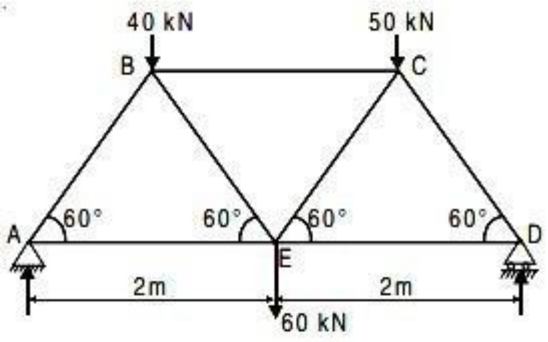
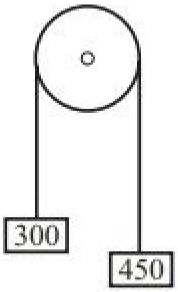
	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE **Winter Examination – 2022** **Course:First Year B. Tech. Branch :Civil/Mechanical/Computer/IT Semester :I** **Subject Code & Name: Engineering Mechanics BTES103** **Max. Marks: 60 Date: Duration: 3 Hr.**		
	Instructions to the Students: 1. *All the questions are compulsory.* 2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.* 3. *Use of non-programmable scientific calculators is allowed.* 4. *Assume suitable data wherever necessary and mention it clearly.*		
		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	**Define: Co-planar forces, Resultant force and Moment of a force.**	**Remember**	**6**
B)	**Determine the resultant of the three forces acting on a hook as shown in figure.**  	**CO2**	**6**
C)	**C.1)State and define free body diagram with suitable examples.** **C2)What are the types of loads and explain with neat sketches**	**Remember**	**6**
Q.2	**Solve Any Two of the following.**		**12**

A)	**Locate the centroid I-section as shown in figure.** 	**CO3**	**6**
B)	**A system of connected flexible cables shown in Figure is supporting two vertical forces 200 N and 250 N at points B and D. Determine the forces in various segments of the cable.** 	**CO2**	**6**
C)	**State and explain Lami's theorem and prove that?**	**Understand**	**6**
Q. 3	**Solve Any Two of the following.**		**12**
A)	**Define: a)Angle of friction b)Coefficient of friction c)Angle of repose d)Cone of friction e)Static friction f)Dynamic friction**	**Remember**	**6**
B)	**A Ladder 5 m long rest on a horizontal ground and leans. against a Smooth vertical wall at an angle of 20° with the vertical. The weight of the ladder is 300N. The ladder is at the Point of Sliding when a man weighing 750N stands on a 1.50m from the foot of the ladder. Calculate the Coefficient of friction between the ladder and the floor.**	**CO2**	**6**

C)	**Determine the forces in all the members of the truss shown in Figure and indicate the magnitude and nature of forces on the diagram of the truss. All inclined members are at 60° to horizontal and length of each member is 2 m.** 	**CO2**	**6**
Q.4	**Solve Any Two of the following.**		**12**
A)	**The equation of motion of a particle moving in a straight line is given by :** $S=18t+3t^2 -2t^3$ **where (s) is in meters and (t) in seconds. Find (1) velocity and acceleration at start, (2) time, when the particle reaches its maximum velocity, and (3) maximum velocity of the particle.**	**CO5**	**6**
B)	**If a particle is projected inside a horizontal tunnel which is 5 meters high with a velocity of 60 m/s, find the angle of projection and the greatest possible range.**	**CO4**	**6**
C)	**Two ships move from a port at the same time Ship A has velocity of 30 kmph and is moving in N 30° W while ship B is moving in south-west direction with a velocity of 40 kmph. Determine the relative velocity of A with respect to B and the distance between them after half an hour.**	**CO4**	**6**
Q. 5	**Solve Any Two of the following.**		**12**
A)	**State and explain in brief D' Alembert's principle.**	**CO5**	**6**
B)	**Two bodies weighing 300 N and 450 N are hung to the ends of a rope passing over an ideal pulley as shown in Figure. find**	**CO5**	**6**

	a)with what acceleration the heavier body comes down? b)What is the tension in the string?  		
C)	A Glass ball is dropped on to a Smooth horizontal floor from which it bounces to a height of 9m, on the second bounce it rises to a height of 6m. from what height the ball was dropped and what is the coefficient of restitution between the glass ball and the floor.	CO5	6
	*** End ***		

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